UNIT-II

TIMERS, PHASE LOCKED LOOP & D-A & A-D CONVERTORS

Introduction:

555 Timer:

→ 555 timer is a timing circuit that can produce accurate & high stable time delays (or) oscillations.

→ 555 timer is available in 8 pin DIP & 14 pin DIP packages.

→ It can be used with supply voltages range in blw +5V to +18V.

→ The below fig. shows the pin diagram of 8 pin DIP package.

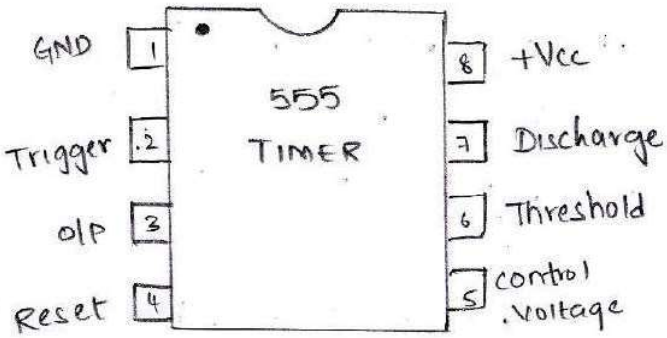


fig: 555 Timer

Features:

1. It can be used with supply voltages over a range in blw +5V to +18V.
2. It is easy to use.
3. It can drive the load upto 200 mA.
4. It is compatable with TTL (Transistor transistor logic) & CMOS (complimentary metal oxide semiconductor).
5. It is used in various applications such as square wave generator, ramp & pulse wave generator, astable & monostable multivibrators.

Functional Diagram:

→ It consists of 2 comparators namely upper comparator & lower comparator that can drive set (S) & reset (R) terminals of a flip flop.

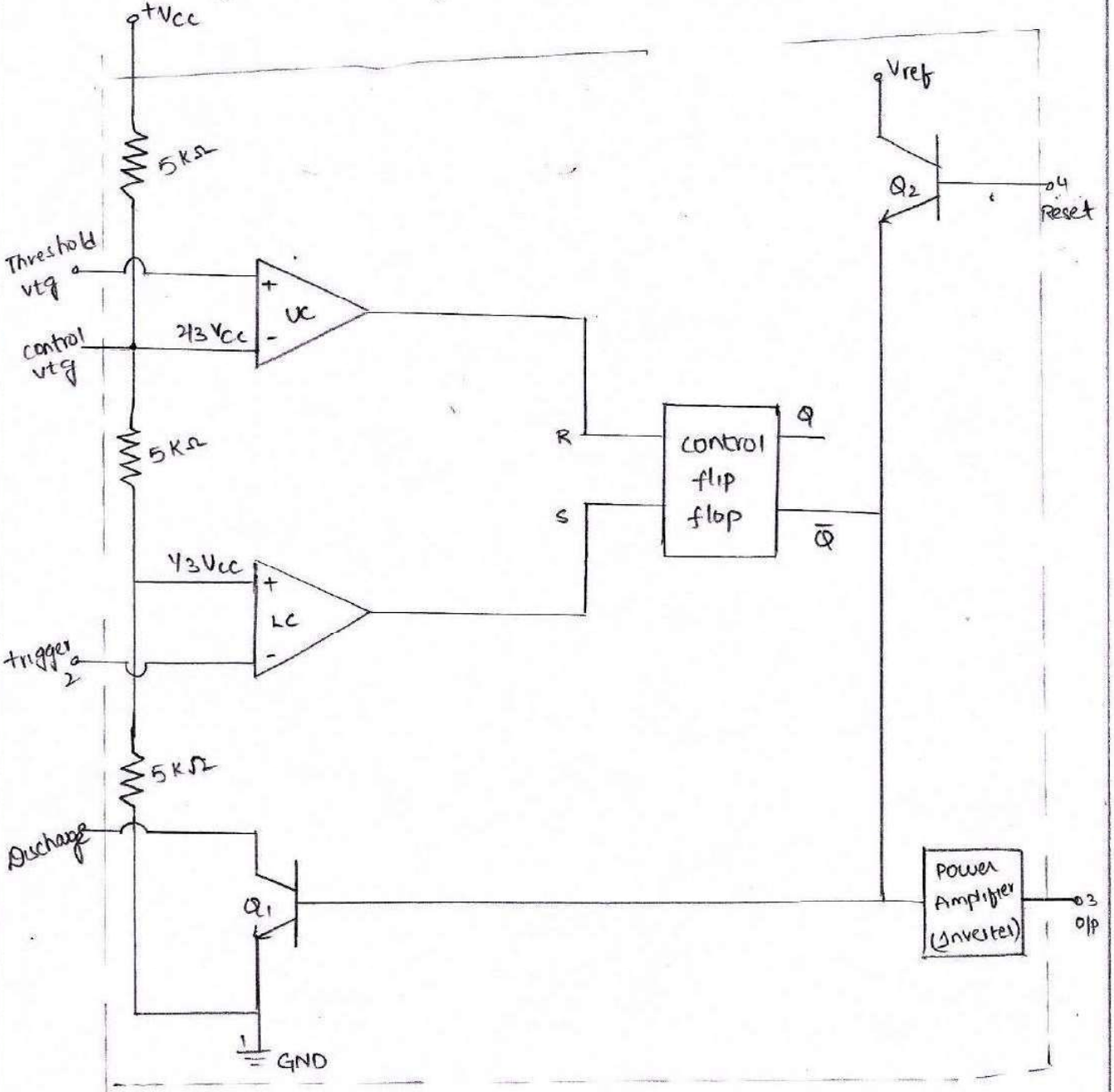
→ These flipflops can control the ON & OFF cycles of the discharge transistor Q_1.
→ It has 3, 5KΩ resistors which acts as potential divider, providing biasing voltages of 2/3 V_{cc} to the upper comparator & 1/3 V_{cc} to the lower comparator where V_{cc} = supply voltage.
→ These voltages are called as reference voltages. These are required to control the timing.
→ The timing can be controlled by externally applying voltage to the control voltage terminal.
→ If no such control voltage is required then the control voltage terminal can be bypaned by a capacitor to ground.
→ Typically the capacitor value is chosen of about 0.1 MF.

Operation:

→ In the stand by state (stable state), the o/p $\bar{Q}$ of the control flipflop is high ($\bar{Q}=1$; $Q=0$). This makes o/p low because of power amplifier can be acts as a inverter.
→ A -ve triggering pulse paves through $\frac{V_{cc}}{3}$, the o/p of the lower comparator goes high & sets the flipflop ($Q=1$; $\bar{Q}=0$)
→ When the threshold voltage at pin 6 paves through 2/3 V_{cc} the o/p of upper comparator goes high & resets the flipflop ($Q=0$; $\bar{Q}=1$)
→ A separate reset terminal is produced to reset the flipflop externally.
→ Normally the reset terminal is not used, if we need it should be connected to +ve supply voltage V_{cc}.
→ The transistor Q_2 acts as buffer to isolate the reset i/p from the flipflop & the transistor Q_1.

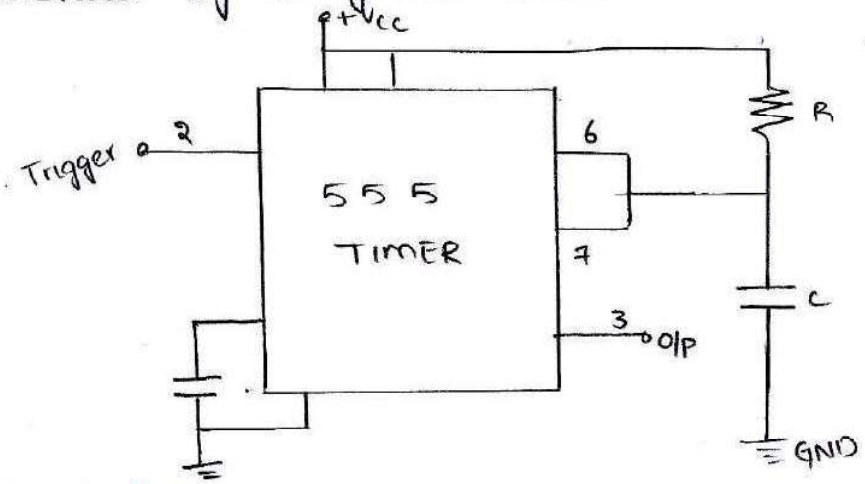
→ The transistor Q_2 is driven by an internal reference voltage V_{ref} obtained from supply voltage V_{cc}.

→ If $\bar{Q}$ is high, the transistor Q_1 is ON due to this it become s/c in b/w discharge pin to ground. Similarly $\bar{Q}$ is low, the transistor Q_1 is OFF & it becomes open circuit in b/w discharge pin to ground.

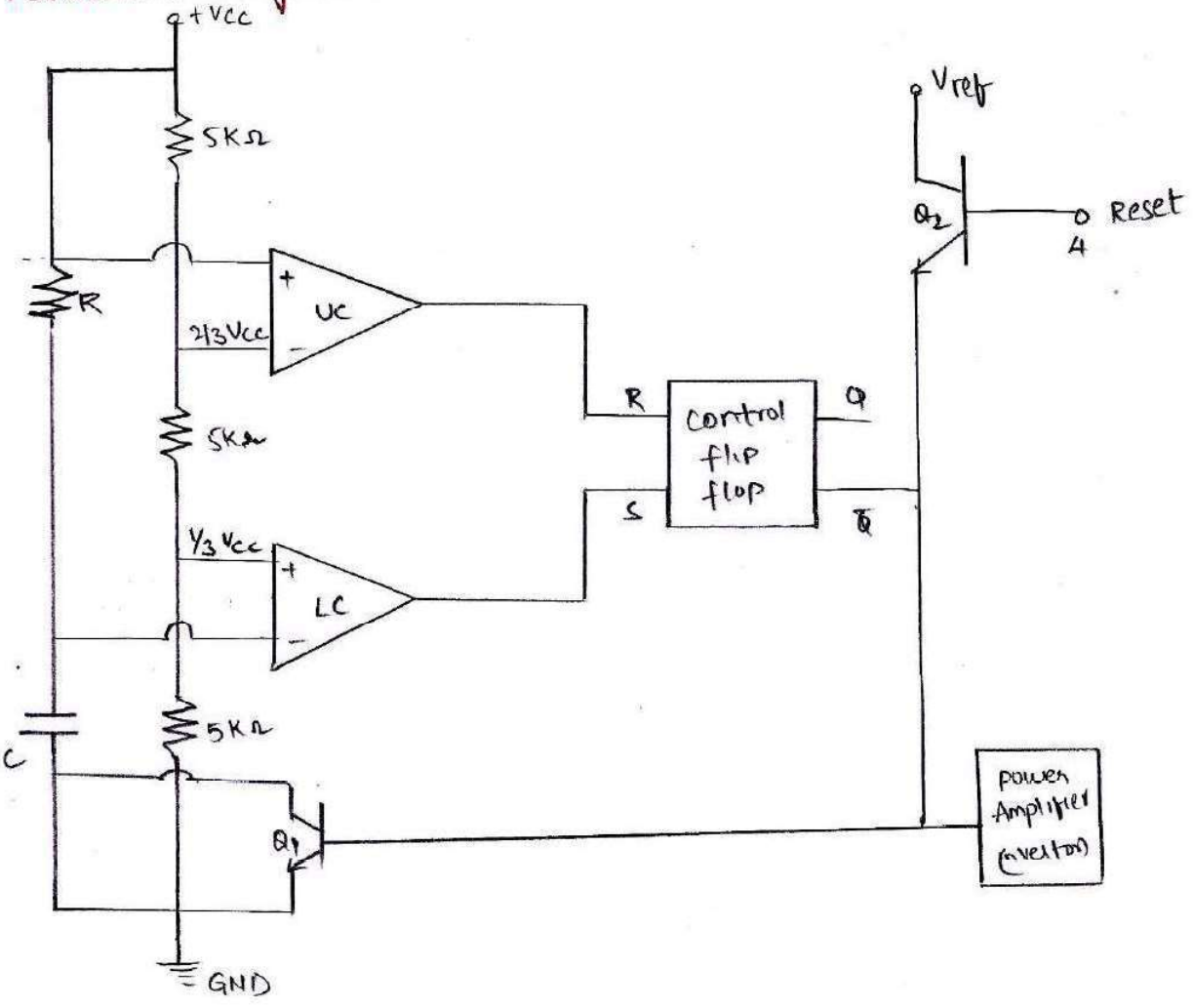


Monostable Multivibrator :-

Monostable multivibrator is a circuit which generates the non-sinusoidal signals. It has one stable state and one quasi stable state. The below fig. shows the monostable multivibrator by using 555 timer.

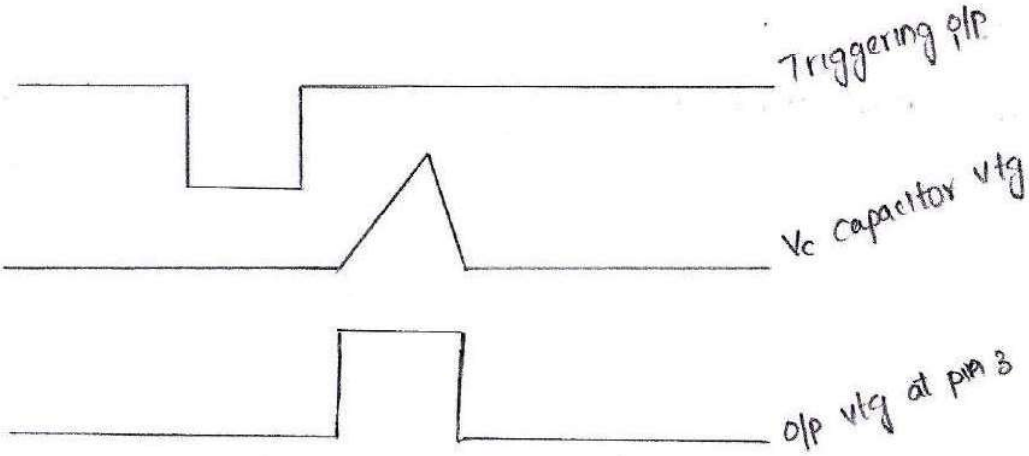


Functional Diagram :-



The above fig. shows the functional block diagram of monostable multivibrator.

1. In the stand by state [stable state] $Q = 0$; $\bar{Q} = 1$. So o/p is low. Under this condition transistor is ON i.e., it becomes short circuit through capacitor. 'C' to the ground.
2. Now the triggering passes through $V_{cc}/3$ at 2nd pin, due to this lower comparator o/p is high. So $Q = 1$, $\bar{Q} = 0$. This makes transistor Q_1 OFF & it becomes a open circuit across the capacitor. so o/p is high.
3. Now, the capacitor takes charging by V_{cc}.
4. After a time period T, the capacitor voltage is just greater than 2/3 V_{cc} and upper comparator o/p is high. so $Q = 0$; $\bar{Q} = 1$.
5. Under this condition, o/p is low & transistor Q_1 goes on there by discharging capacitor 'C' rapidly to ground.
6. The corresponding o/p waveforms of monostable multivibrator is shown in fig.



Analysis of Time Constant :

The capacitor voltage across the capacitor is given by

$$V_c = V_{cc}\left(1 - e^{-t/RC}\right) \rightarrow (1)$$

At $t = T$, the capacitor charges by $V_c = 2/3\ V_{cc}$ → (2)

sub. eq(2) in eq(1)

$$2/3\ V_{cc} = V_{cc} - V_{cc}\, e^{-t/RC}$$

$$2/3\ V_{cc} + V_{cc}\, e^{-T/RC} = V_{cc}$$

$$e^{-T/RC} = \frac{V_{cc} - 2/3\ V_{cc}}{V_{cc}}$$

Apply log on b.s

$$\ln[a/b] = -\ln[b/a]$$

$$\ln\left[e^{-T/RC}\right] = \ln\left[\frac{V_{cc} - 2/3\ V_{cc}}{V_{cc}}\right]$$

$$+\frac{T}{RC} = +\ln\left[\frac{V_{cc}}{V_{cc} - 2/3\ V_{cc}}\right]$$

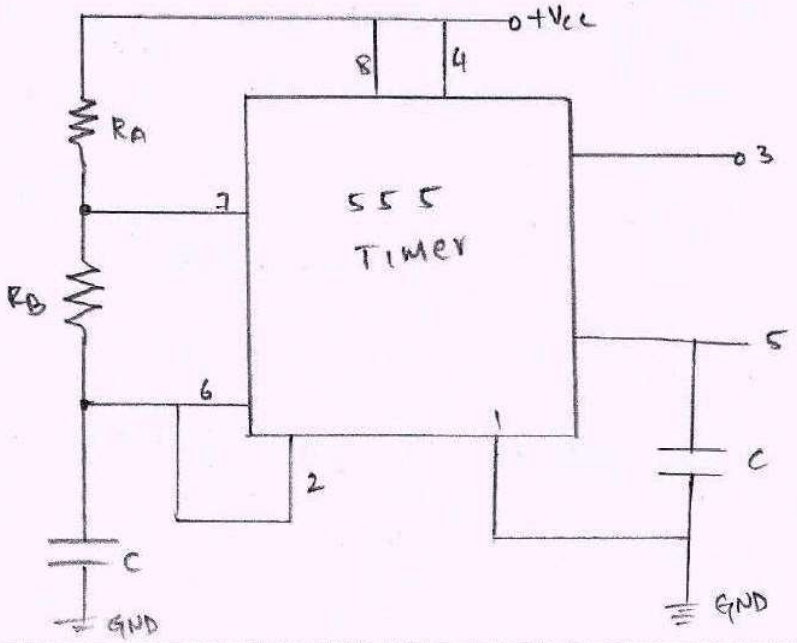
$$T = \ln RC(3)$$

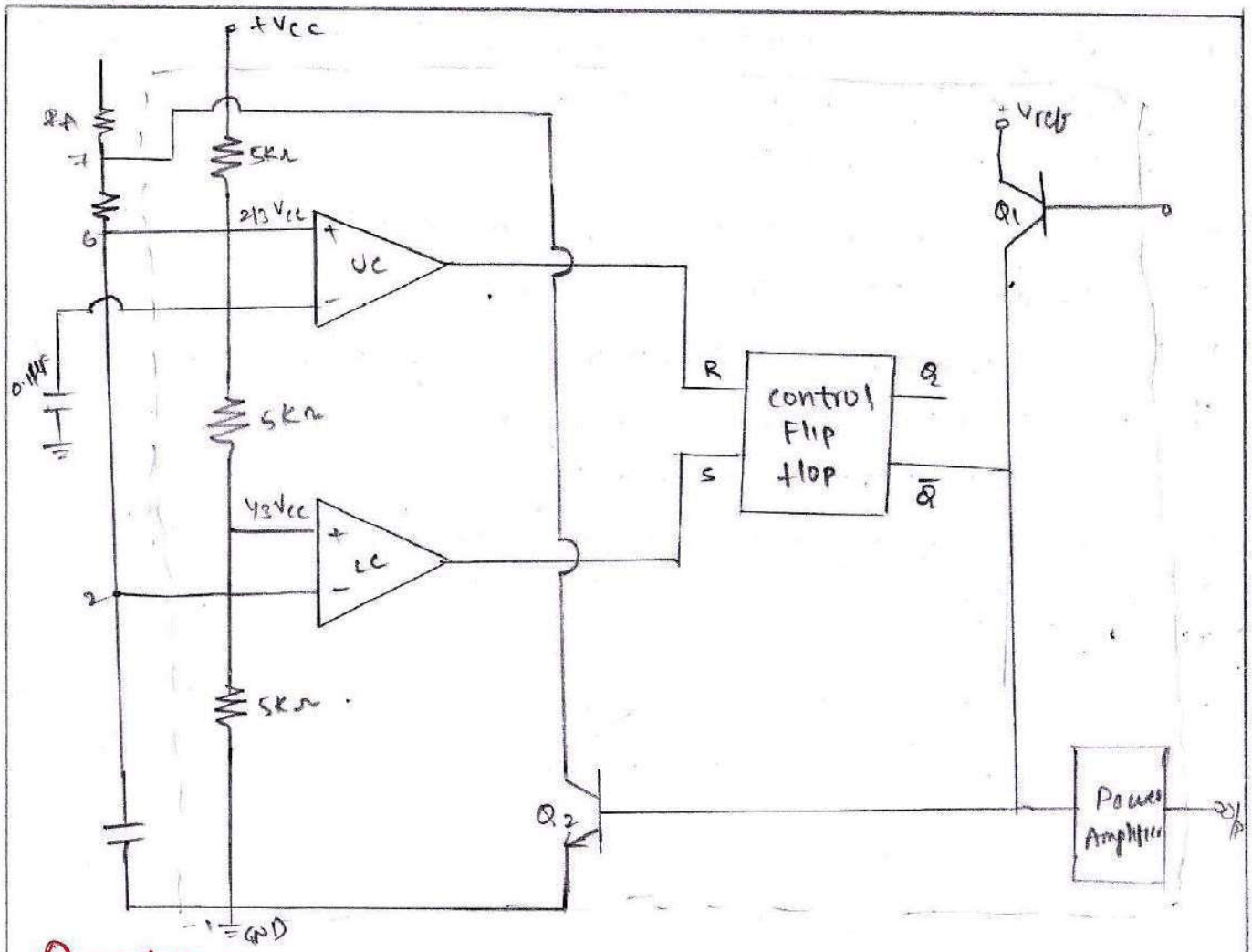
$$\therefore T = 1.1\ RC$$

Applications :

1. Pulse width generator
2. Water level control.

Asstable Multivibrator :





Operation :

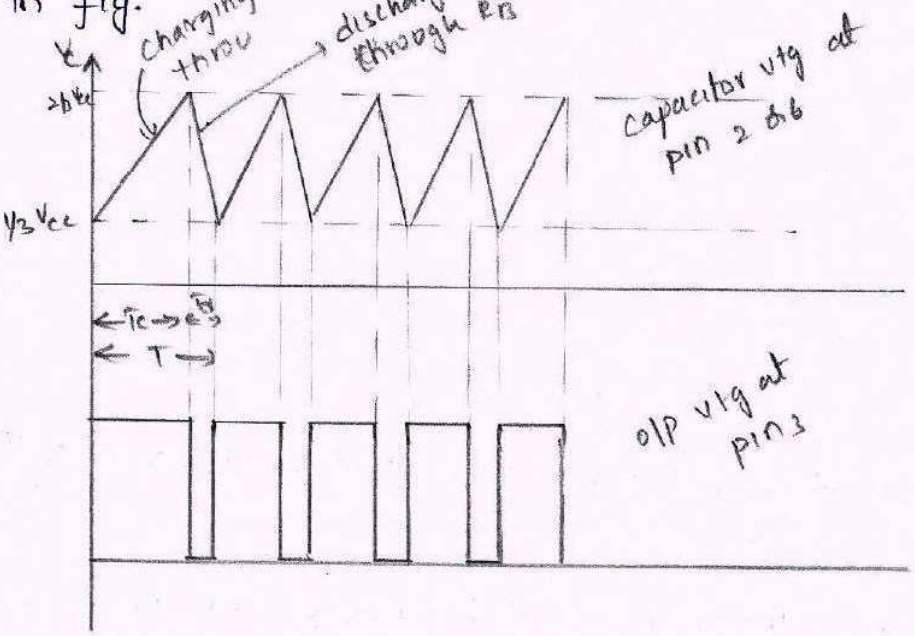
1. Asstable multivibrator has no stable states. It has 2 quasi stable states.
2. The astable multivibrator circuit by using 555 timer is shown in above fig.
3. Comparing with monostable multivibrator, the timing resistor is now split into 2 sections i.e., R_A & R_B.
4. The discharging transistor Q_1 is connected in blw R_A & R_B.
5. When supply voltage is connected, the external capacitor gets charging through R_A & R_B resistors.
6. When the charging voltage reaches the 1/3 Vcc voltage then lower comparator goes high. Due to this $R=0$; $S=1$; $\bar{Q}=0$; $Q=1$ and therefore o/p is high.

7. Similarly when Charging voltage reaches 2/3 Vcc then upper comparator o/p goes high. Due to this R=1 ; S=0 ; $\bar{Q}$=1 ; Q=0 and the o/p becomes low.

8. When $\bar{Q}$=1 , the discharging transistor Q_2 is ON & it makes short circuited across the capacitor.

9. So the capacitor gets discharging through R_B resistor towards the ground. The capacitor discharging voltage reaches 1/3 Vcc & again lower comparator o/p goes high.

10. The corresponding waveforms of Astable multivibrator is shown in fig.



Analysis of time constant :-

$$V_c = V_{cc}(1-e^{-t/RC}) \rightarrow (1)$$

at $t=t_1$, $V_c = 1/3\,V_{cc}$

$$1/3\,V_{cc} = V_{cc} - V_{cc}\,e^{-t_1/RC}$$

$$V_{cc}\,e^{-t_1/RC} = V_{cc} - 1/3\,V_{cc}$$

$$e^{-t_1/RC} = \frac{V_{cc} - 1/3\,V_{cc}}{V_{cc}}$$

Taking log on b/s

$$-\frac{t_1}{RC} = \ln\left[\frac{V_{CC} - 1/3\,V_{CC}}{V_{CC}}\right]$$

$$-\frac{t_1}{RC} = -\ln\left[\frac{V_{CC}}{V_{CC} - 1/3\,V_{CC}}\right]$$

$$t_1 = RC \ln\left[\frac{1}{1-1/3}\right]$$

$$t_1 = RC \ln(1.5)$$

$$\therefore t_1 = 0.405\,RC$$

→ When the capacitor charges from 1/3 V_{CC} to 2/3 V_{CC}.

at $t = t_2$; $V_{ce} = 2/3\,V_{CC}$

$$2/3\,V_{CC} = V_{CC} - V_{CC}\,e^{-t_2/RC}$$

$$V_{CC}\,e^{-t_2/RC} = V_{CC} - 2/3\,V_{CC}$$

$$e^{-t_2/RC} = \frac{V_{CC} - 2/3\,V_{CC}}{V_{CC}}$$

Taking log on b/s

$$-t_2/RC = \ln\left[\frac{V_{CC} - 2/3\,V_{CC}}{V_{CC}}\right]$$

$$t_2 = \ln RC\left[\frac{V_{CC}}{V_{CC} - 2/3\,V_{CC}}\right]$$

$$t_2 = RC \ln(3)$$

$$\therefore t_2 = 1.1\,RC$$

$$T_C = t_2 - t_1$$

$$= 1.1 - 0.405$$

$$= 0.695\,RC$$

$$\therefore T_C = 0.695\,(R_A + R_B)\,C$$

→ The capacitor takes discharging from 2/3 Vcc to 1/3 Vcc

$$1/3\ V_{cc} = 2/3\ V_{cc}\ e^{-t/RC}$$

$$e^{-t/RC} = \frac{1/3\ V_{cc}}{2/3\ V_{cc}}$$

$$e^{-t/RC} = 1/2$$

$$-t/RC = \ln(1/2)$$

$$t/RC = \ln(2)$$

$$t = RC\ln(2)$$

$$\therefore T_d = 0.69 R_B C$$

→ Total time constant $T = T_C + T_D$

$$= 0.69(R_A + R_B)C + 0.69 R_B C$$

$$= 0.69 R_A C + 0.69 R_B C + 0.69 R_B C$$

$$\therefore T = 0.69 (R_A + 2R_B) C$$

$$f = 1/T$$

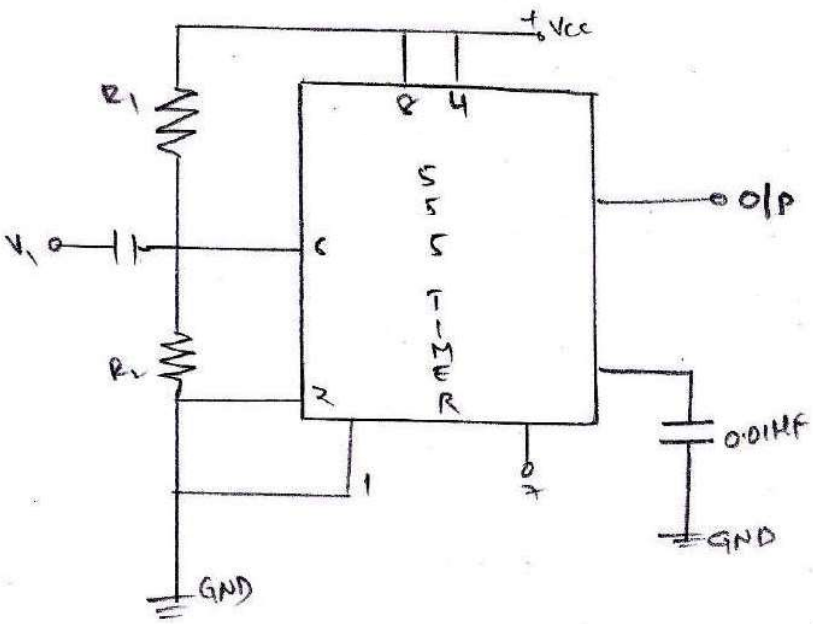
$$= \frac{1}{0.69(R_A + 2R_B)C}$$

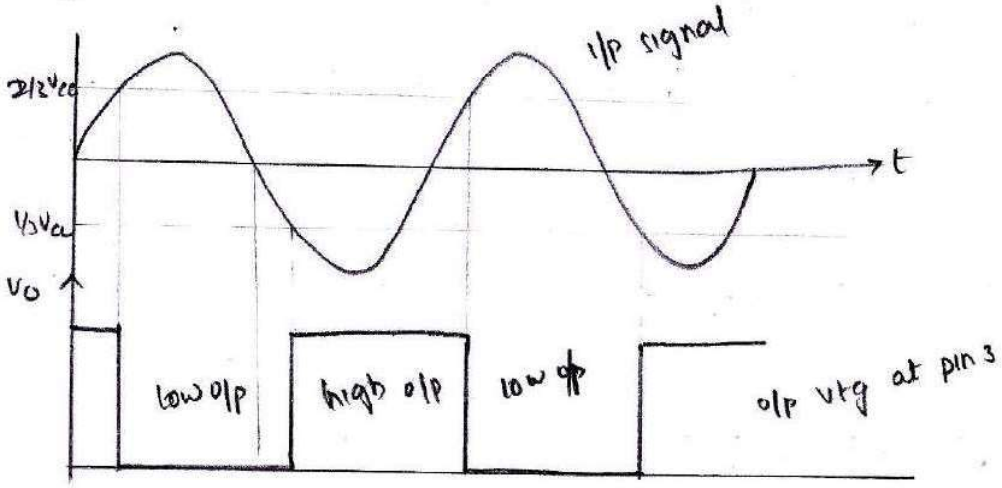
$$\therefore f = \frac{1.449}{(R_A + 2R_B)C}$$

Applications :-

1. Frequency shift key (FSK)
2. Pulse position modulator

Schmitt Trigger :-





→ Here 2 internal comparators are connected together an externally biased through R_1 & R_2 resistors.

→ Since upper comparators will be ON (or) o/p is high when the given i/p signal reaches the 2/3 Vcc.

→ Similarly lower comparator o/p is high when the given i/p signal reaches the 1/3 Vcc.

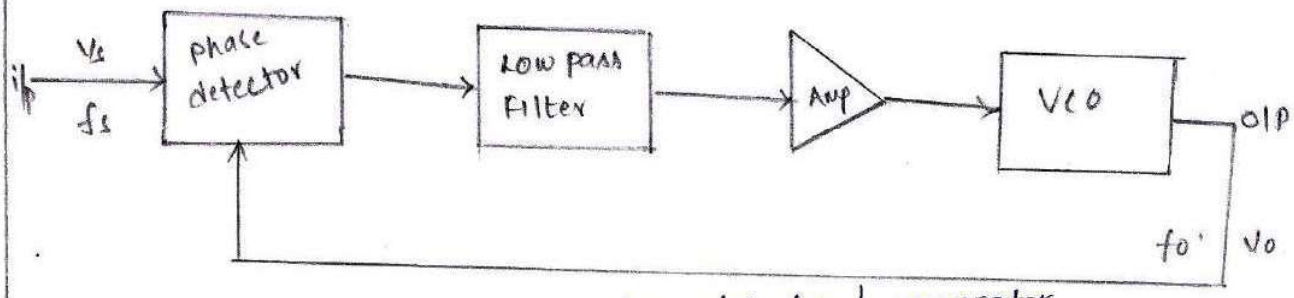
→ Due to this the o/p of the circuit is changing from one state to another state. i.e., when uc is high, o/p is low when lc is high, o/p is high (R=0 ; S=1 ; ϕ=1 ; $\bar{\phi}$=0).

Phase Locked Loop :-

Introduction :

1. PLL is a phase locked loop. It is a closed loop circuit & its o/p frequency & o/p phase (ϕ) to be locked.
2. The PLL is an important building block of linear systems.
3. The PLL was used in 1930. At the time PLL has many features. So PLL circuits was very costly.
4. However, after the development of integrated technology, the cost of PLL has reduced.
5. Hence we observed that PLL has become one of the fundamental building block in electronic technology.
6. The PLL principle is used in FM demodulation, FSK demodulation, motor speed control, frequency multiplication & division etc.
7. The PLL is available in single package. The example of PLL is 565 IC.

Block Diagram of PLL :



It consists of 4 blocks ; 1. phase detector / comparator
2. Low pass filter
3. Amplifier
4. VCO (voltage controlled oscillator)

1. Phase Detector / comparator :

1. When i/p signal V_s at frequency f_s is applied to the phase detector & it compares the phase or frequency of incoming signal to that of the o/p of VCO.
2. The phase detector compares the 2 i/p signals & produce an voltage.
3. Phase detector basically acts as an multiplier, so it produces the sum $(f_s + f_o)$ & difference $(f_s - f_o)$ components at its o/p.

2. Low pass Filter :

The low pass filter used to remove high frequency signals i.e., coming from phase detector. It passes only low frequency signals i.e., the difference of two i/p signal $(f_s - f_o)$.

3. Amplifier :

The amplifier is used to amplifies the difference of frequency signal & the amplified signal is given to the voltage controlled oscillator.

4. Voltage Controlled Oscillator :

1. VCO is a frequency running multivibrator and operates at a set frequency f_o called free running frequency.
2. This frequency is determined by an external timing capacitor and an external resistor.
3. It can be shifted to either side by applying a dc control voltage V_c.
4. The frequency derivation is directly proportional to the dc ~~current~~ control voltage and it is called VCO.

5. The VCO frequency f_o is compared with the i/p frequency f_i by the phase detector and it is adjusted continuously until it is equal to the i/p frequency f_s.

$$f_o = f_s$$

6. The signal V_c shifts the VCO frequency in a direction to reduce the frequency difference b/w f_s and f_o.

7. Once this action starts, we say that the signal is in the capture range.

8. The circuit is then said to be locked. Once locked, the o/p frequency f_o of VCO is identical (same) to f_s except for a finite phase ϕ. Thus, a PLL goes through 3 stages.

1. Free Running state :

In this state, there is no control on VCO o/p frequency f_o.

2. Lock Range :

In this state when f_o is exactly equal to f_i, the PLL is said to be phase locked. Once locked $f_o = f_i$ except for a finite ϕ.

3. Capture Range :

In this state, the comparision of f_o and f_i begins. The control voltage V_c starts adjusting f_o to bring it closer to f_i. The LPF controls the capture range.